Danium Shahnowaz Syed

Applied Cybersecurity

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Google Scholar | LinkedIn

Summary

Passionate Computer Engineer and Erasmus Mundus Scholar. Solid experience in DevOps, System Administration, IIoT and privacy/security research. Holding a double Master's degree in Cybersecurity and a CCNA certification.

Education

Erasmus Mundus Master's in Applied Cybersecurity (CyberMACS), Kadir Has University & SRH Berlin University of Applied Sciences – Istanbul & Berlin — Sept 2024 – Sept 2026

- Selected among the top 28 fully funded students from over 1,000 applications
- Received a fully funded scholarship given to the top applicants
- GPA: 3.43/4.00 (KHAS), 1.8 (SRH, German scale)

BSc in Electrical and Electronics Engineering, American International University-Bangladesh (AIUB) – Dhaka, Bangladesh — Sept 2019 – Sept 2023

- Received multiple awards and studied on a full scholarship since admission
- Graduated top of class with a CGPA of 3.98/4.00 (1.03 German scale)
- Received Best Project Award for Thesis/Capstone Project

IGCSE and A-Levels in Pearson Edexcel British Curriculum, Mount Royal Academy – Bangladesh — June 2017 – June 2019

Additional Academic Training

Guest Coursework, Technical University of Darmstadt – Darmstadt, Germany — In progress

- Cryptography on Hardware: Implementing cryptographic functions, including AES, on FPGAs using VHDL.
- Physical Attacks & Countermeasures: Performing side-channel attacks on FPGA-based cryptographic implementations and studying countermeasures.

Experience

Cybersecurity Intern, Cyberforce – Istanbul, Turkey — June 2025 – Sept 2025

- Developed internal tools for security testing of LLM/ML models from PyTorch, TensorFlow, Keras, Sklearn, and XGBoost, detecting insecure serialization formats inside Docker containers and malware-corrupted models to prevent model serialization attacks
- Conducted on-site and offline penetration testing of ML and LLM models; completed all levels of the Gandalf and Prompt Airlines CTFs
- Conducted security research on Telecom Networks and IoT networks, using the HackkRF One SDR

System Administrator, SECOM Technologies Ltd. – Dhaka, Bangladesh — Mar 2023 – Aug 2024

- Managed Linux, macOS, and Windows virtual machines supporting internal business applications
- Administered Proxmox virtualization infrastructure and used Ansible playbooks

ICT and DevOps Intern, Grameenphone (Telenor) – Dhaka, Bangladesh Jan 2023 – June 2023

- Automated ICT workflows and built an internal tool to monitor monthly product revenues across 8 products, presenting them on a dashboard
- Worked on the system architecture formation of Grameenphone Alo, a proprietary IoT platform
- Extensively worked on hybrid-cloud architectures and Kubernetes, and on secure CI/CD deployment through Jenkins, Docker, and Terraform
- Conducted on-site surveys of Grameenphone datacenters

IT Educator, Dignitus International School – Narayanganj, Bangladesh July 2017 – July 2019

- Educated students from grade 8 to 10, with multiple A* achievements in taught subjects for board exams
- Maintained lesson plans and liaised with the international education board Edexcel

Publications

Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains Nov 2026
Danium S. Syed
(Submitted to SECITC (Springer LNCS) conference proceedings)

Analyzing Political Sentiment on Twitter with Zero-Shot Large Language Models Oct 2026
R. Ayoub, *Danium S. Syed*, E. O. Njoku
(Accepted in Proc. IACyC 2026)

Explainable Botnet Detection Using Flow-Based Random Forest Classification and OSINT Validation Apr 2026
A. Razan, E. O. Njoku, *Danium S. Syed*, N. Nachevski
(Web Proc. CIIT 2026)

Reputation and Energy-Aware Dynamic Hybrid Consensus (READ-HC) Model for IIoT July 2025
S. Mitra, V. S. Sable, *Danium S. Syed*
10.31585/jbba-8-2-(3)2025 (The Journal of The British Blockchain Association)

Design and Feasibility Analysis of a LoRa Based Communication System for Disaster Management Aug 2024
Danium S. Syed, S. B. Hossain, A. Islam, T. A. I. Jim, M. Saniat, R. Zishan
10.53799/ajse.v23i2.1055 (AIUB Journal of Science and Engineering (AJSE), 23, 135-144)

Projects

Master's Thesis: Security Analysis of Machine Learning Artifacts in Containerized AI Supply Chains Jan 2026 – present

- Conducted security analysis of containerized ML systems to measure prevalence and distribution of embedded model artifacts across Docker ecosystems
- Identified security risks in ML supply chains by statically analyzing model files (PyTorch, TensorFlow, ONNX, Pickle) for unsafe serialization and code execution vulnerabilities
- Built an automated pipeline to extract container filesystems and scan ML artifacts, generating empirical insights into real-world deployment risks

Linux Home Server (HomeLab) March 2020 - Ongoing

- Built from scratch and maintains a Debian-based home server for security research, home automation, and media consumption
- Follows DevOps practices and maintains high uptime
- Technology stack: WireGuard, Docker Compose, Proxmox

Bachelor's Capstone: Long-Range Off-Grid Communication System Deployed with Amphibious Rovers Jan 2023 – Sept 2023

- Built a mesh communication system using LoRa modulation for deployment in disaster scenarios via amphibious rovers
- Provides text messaging and location-based communication services for affected users in disaster zones

More projects: daniumsyed.github.io/projects

Honors and Awards

- 5th Place (All Solves), CYBERUS Embedded Systems CTF Competition — Lorient, France (2026)
- 2nd Place (All Solves), CyberMACS Winter School CTF Competition — Istanbul, Turkey (2025)
- Erasmus Mundus Full Scholarship, EMJM in Applied Cybersecurity — Turkey/Germany (2024)
- Dean's Best Project Award, AIUB — Dhaka, Bangladesh (2023)
- Dean's List Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Freedom Fighter's Scholarship, AIUB — Dhaka, Bangladesh (2019)
- Edexcel High Achiever's Award, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)
- Featured in The Daily Star and The Daily Sun, IGCSE/Grade 10 — Dhaka, Bangladesh (2017)

Certifications

- Building Resilience: Navigating Vulnerability, Risks, and Continuity in a Dynamic World — CyberMACS (2026)
- Cisco Certified Network Associate (CCNA), CSCO14934967 — Cisco (2025)
- The Bits and Bytes of Computer Networking — Google/Cisco, Coursera (2020)

Skills

DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

System Administration: Linux, Networking, Virtualization, Containers, Git

Privacy & AI Security: Differential Privacy, Agentic AI, PyTorch, TensorFlow, vLLM/Ollama

Security Research: Vulnerability Analysis, Threat Modeling (MITRE ATT&CK), Penetration Testing (Hardware and Software), Incident Response

Programming/Scripting: Python, JavaScript, Bash, LaTeX, C/C++, SQL, HTML/CSS

Languages: English (IELTS 8.0), Bengali (Native), Turkish (A1), German (A1, ongoing studies)